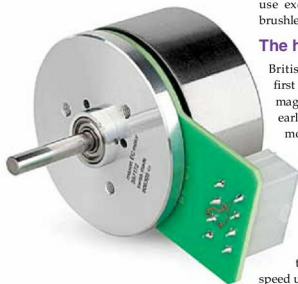


Brushless DC Motors Replacing Brush Motors

Brushless DC motors, once too expensive, cost much less today, making them a popular choice for many different applications. Find out what makes brushless DC motors better than brush motors, trends in their design and applications



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Brushless direct-current (BLDC) motors have received considerable attention throughout the industrial world since the early 1970s. These have become widespread and are available in all shapes and sizes from large-scale industrial models to small motors for light applications. BLDC motors are most commonly used in easy-to-drive, variable-speed and long-life applications.

Although brushless DC motors and conventional DC motors are similar in their static characteristics, these actually remarkably differ in some aspects. As the name implies, a brushless DC motor does not operate by using brushes. Rather, it operates with a controller via electronic commutation. Computer disk drives and small fans

use exclusively permanent-magnet, brushless DC motors.

The history

British inventor Michael Faraday first tried out the idea of electromagnetic induction motor in the early 18th century. By 1828, DC motor was introduced with three main components: stator, rotor and commutator. Later, in 1886, Frank Julian Sprague introduced the first practical brushed DC motor (containing a permanent magnet and a moving coil within the magnet) that was capable of constant speed under variable loads.

Over time, elimination of commutator and brushes made room for the inception of electronic commutation and brushless DC motors (containing a moving magnet unit and a permanent coil assembly). Although brushless DC motor was very expensive when first introduced, the advancements in design and materials drastically lowered its cost, making it a popular choice for many different applications.

What is inside?

Typically, a BLDC motor consists of a permanent magnet rotor with a three-phase stator winding. Stator windings are connected to an electronic commutation system—an alternative to the mechanical commutation system utilised by a brush DC motor.

The rotor in BLDC motor consists of an even number of permanent magnets. The number of magnetic poles in the rotor also affects the step size

and torque ripple of the motor. More poles give smaller steps and less torque ripple.

Electromagnets within the motor remain stationary along with the armature, while the encased permanent magnets rotate, generating torque. As the orientation of rotor is crucial for successful operation of brushless DC motor, Hall Effect sensors (transducers that vary their output voltage in response to changes in magnetic field) are usually used to detect the exact position of rotor. Typically, three Hall sensors are used to detect rotor position. Commutation is based on these sensor inputs.

BLDC motor control

An efficient BLDC motor controller (also known as driver or electronic speed controller) is essential for speed, direction and brake control of BLDC

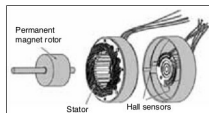


Fig. 1: Disassembled view of a basic BLDC motor

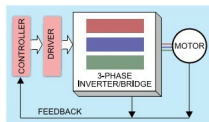


Fig. 2: Basic block diagram of BLDC motor